

Double Cross product: (Practice)

- let's show (page 24 of your notes) how to obtain:

$$\vec{a} \times (\vec{b} \times \vec{c}) = \vec{b}(\vec{a} \cdot \vec{c}) - \vec{c}(\vec{a} \cdot \vec{b})$$

- We know:

$$\vec{b} \times \vec{c} = \begin{vmatrix} \hat{e}_x & \hat{e}_y & \hat{e}_z \\ b_x & b_y & b_z \\ c_x & c_y & c_z \end{vmatrix} = (b_y c_z - b_z c_y) \hat{e}_x - (b_x c_z - b_z c_x) \hat{e}_y + (b_x c_y - b_y c_x) \hat{e}_z$$

let's call these

$$\alpha \equiv (b_y c_z - b_z c_y)$$

$$\beta \equiv -(b_x c_z - b_z c_x)$$

$$\gamma \equiv (b_x c_y - b_y c_x)$$

then

$$\begin{aligned} \vec{b} \times \vec{c} &= \alpha \hat{e}_x + \beta \hat{e}_y + \gamma \hat{e}_z = (\alpha, \beta, \gamma) \\ &= \begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix} \leftarrow (\text{Matrix notation}) \end{aligned}$$

Thus,

$$\begin{aligned} \vec{a} \times (\vec{b} \times \vec{c}) &= \begin{vmatrix} \hat{e}_x & \hat{e}_y & \hat{e}_z \\ a_x & a_y & a_z \\ \alpha & \beta & \gamma \end{vmatrix} \\ &= (a_y \gamma - a_z \beta) \hat{e}_x - (a_x \gamma - a_z \alpha) \hat{e}_y \\ &\quad + (a_x \beta - a_y \alpha) \hat{e}_z \end{aligned}$$

$$(i) (a_y \gamma - a_z \beta) = a_y (b_x c_y - b_y c_x) + a_z (b_x c_z - b_z c_x)$$

- combine b_x and c_x terms:

$$= b_x (a_y c_y + a_z c_z) - c_x (a_y b_y + a_z b_z)$$

$$\text{add } + b_x a_x c_x - b_x a_x c_x = 0$$

to above:

$$= b_x (a_y c_y + a_z c_z) + b_x a_x c_x$$

$$\begin{aligned}
 & - C_x (a_y b_y + a_z b_z) - C_x a_x b_x \\
 & = b_x (a_x c_x + a_y c_y + a_z c_z) \\
 & \quad - C_x (a_x b_x + a_y b_y + a_z b_z)
 \end{aligned}$$

Note: $\vec{a} \cdot \vec{c} = (a_x c_x + a_y c_y + a_z c_z)$
 $\vec{a} \cdot \vec{b} = (a_x b_x + a_y b_y + a_z b_z)$
 therefore

$$= b_x (\vec{a} \cdot \vec{c}) - C_x (\vec{a} \cdot \vec{b})$$

(ii) $-(a_x \gamma - a_z \alpha) = a_z \alpha - a_x \gamma$

• Do the exact same, you'll find

$$= b_y (\vec{a} \cdot \vec{c}) - C_y (\vec{a} \cdot \vec{b})$$

(iii) $a_x \beta - a_y \alpha$:

Exactly the same:

$$= b_z (\vec{a} \cdot \vec{c}) - C_z (\vec{a} \cdot \vec{b})$$

Plugging this in:

$$\begin{aligned}
 \vec{a} \times (\vec{b} \times \vec{c}) &= (b_x (\vec{a} \cdot \vec{c}) - C_x (\vec{a} \cdot \vec{b})) \hat{e}_x \\
 &\quad + (b_y (\vec{a} \cdot \vec{c}) - C_y (\vec{a} \cdot \vec{b})) \hat{e}_y \\
 &\quad + (b_z (\vec{a} \cdot \vec{c}) - C_z (\vec{a} \cdot \vec{b})) \hat{e}_z
 \end{aligned}$$

• Collect $(\vec{a} \cdot \vec{b})$ and $(\vec{a} \cdot \vec{c})$ terms

$$(\vec{a} \cdot \vec{c}) (b_x \hat{e}_x + b_y \hat{e}_y + b_z \hat{e}_z) - (\vec{a} \cdot \vec{b}) (C_x \hat{e}_x + C_y \hat{e}_y + C_z \hat{e}_z)$$

$\underbrace{\hspace{10em}}_{\vec{b}} \qquad \underbrace{\hspace{10em}}_{\vec{c}}$

Thus:

$$\vec{a} \times (\vec{b} \times \vec{c}) = \vec{b}(\vec{a} \cdot \vec{c}) - \vec{c}(\vec{a} \cdot \vec{b})$$

Problem sheet 2:

- problems 2/3 can be done together:

→ Note for both theres an implicit scalar product:

$$2) (\vec{a} \times \vec{b})^2 = (\vec{a} \times \vec{b}) \cdot (\vec{a} \times \vec{b})$$

$$3) \vec{c} \cdot (\vec{a} \times (\vec{b} \times \vec{c})) + (\vec{c} \times \vec{a}) \cdot (\vec{c} \times \vec{b})$$

* We know

$$\star \vec{a} \cdot (\vec{b} \times \vec{c}) = \vec{b} \cdot (\vec{c} \times \vec{a}) = \vec{c} \cdot (\vec{a} \times \vec{b})$$

Lets compute:

$$(\vec{\alpha} \times \vec{\beta}) \cdot (\vec{\gamma} \times \vec{\delta})$$

i) call $\vec{u} = (\vec{\alpha} \times \vec{\beta})$, then this becomes

$$\vec{u} \cdot (\vec{\gamma} \times \vec{\delta}) = \vec{\gamma} \cdot (\vec{\delta} \times \vec{u})$$

↑

Here we use $\star$

$$= \vec{\gamma} \cdot (\vec{\delta} \times (\vec{\alpha} \times \vec{\beta}))$$

we found this above:

$$\vec{\delta} \times \vec{\alpha} \times \vec{\beta} = \vec{\alpha}(\vec{\delta} \cdot \vec{\beta}) - \vec{\beta}(\vec{\delta} \cdot \vec{\alpha})$$

Plugging this in:

$$= \vec{\gamma} \cdot (\vec{\alpha}(\vec{\delta} \cdot \vec{\beta}) - \vec{\beta}(\vec{\delta} \cdot \vec{\alpha}))$$

$$= (\vec{\gamma} \cdot \vec{\alpha})(\vec{\delta} \cdot \vec{\beta}) - (\vec{\gamma} \cdot \vec{\beta})(\vec{\delta} \cdot \vec{\alpha})$$

Thus:

$$(\vec{\alpha} \times \vec{\beta}) \cdot (\vec{\gamma} \times \vec{\delta}) = (\vec{\alpha} \cdot \vec{\gamma})(\vec{\beta} \cdot \vec{\delta}) - (\vec{\alpha} \cdot \vec{\delta})(\vec{\beta} \cdot \vec{\gamma})$$

OK ... we can use this to solve

okay, so now we can use this to solve both problems

$$(2) (\vec{a} \times \vec{b}) \cdot (\vec{a} \times \vec{b})$$

here

$$\vec{\beta} = \vec{b}, \vec{\gamma} = \vec{a}, \vec{\delta} = \vec{b}$$

$$\begin{aligned} (\vec{a} \times \vec{b})^2 &= (\vec{a} \cdot \vec{a})(\vec{b} \cdot \vec{b}) - (\vec{a} \cdot \vec{b})(\vec{b} \cdot \vec{a}) \\ &= |\vec{a}|^2 |\vec{b}|^2 - (\vec{a} \cdot \vec{b})^2 \end{aligned}$$

$$(3) \underbrace{\vec{c} \cdot (\vec{a} \times (\vec{b} \times \vec{c}))}_{\text{term 1}} + \underbrace{(\vec{c} \times \vec{a}) \cdot (\vec{c} \times \vec{b})}_{\text{term 2}}$$

Term 1: $(\vec{a} \times (\vec{b} \times \vec{c})) = \vec{b}(\vec{a} \cdot \vec{c}) - \vec{c}(\vec{a} \cdot \vec{b})$
from above, thus

$$\begin{aligned} \vec{c} \cdot (\vec{a} \times (\vec{b} \times \vec{c})) &= \vec{c} \cdot (\vec{b}(\vec{a} \cdot \vec{c}) - \vec{c}(\vec{a} \cdot \vec{b})) \\ &= \underbrace{(\vec{c} \cdot \vec{b})(\vec{a} \cdot \vec{c})} - \underbrace{(\vec{c} \cdot \vec{c})(\vec{a} \cdot \vec{b})} \end{aligned}$$

Term 2: $(\vec{c} \times \vec{a}) \cdot (\vec{c} \times \vec{b})$

Here $\alpha = c, \gamma = c$
 $\beta = a, \delta = b$

so, using our formula:

$$\underline{(\vec{c} \times \vec{a}) \cdot (\vec{c} \times \vec{b}) = (\vec{c} \cdot \vec{c})(\vec{a} \cdot \vec{b}) - (\vec{c} \cdot \vec{b})(\vec{a} \cdot \vec{c})}$$

Plugging both in:

$$\begin{aligned} &\vec{c} \cdot (\vec{a} \times (\vec{b} \times \vec{c})) + (\vec{c} \times \vec{a}) \cdot (\vec{c} \times \vec{b}) \\ &= \underbrace{(\vec{c} \cdot \vec{b})(\vec{a} \cdot \vec{c})} - \underbrace{(\vec{c} \cdot \vec{c})(\vec{a} \cdot \vec{b})} + \underbrace{(\vec{c} \cdot \vec{c})(\vec{a} \cdot \vec{b})} - \underbrace{(\vec{c} \cdot \vec{b})(\vec{a} \cdot \vec{c})} \end{aligned}$$

$\rightarrow$ Terms cancel

$$= 0$$

Extra identity:

Jacobi: $A \times (B \times C) + B \times (C \times A) + C \times (A \times B) = 0$

